Name: ______________________

Class: ______________________

Date: ______________________

Mr. Rawson • GCSE Physics

RP 15: Resistance of a Wire

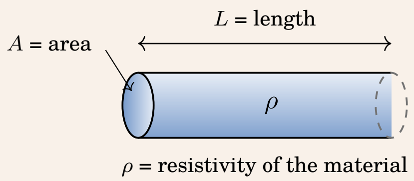
Key concept

You cannot measure resistance directly with the meters on your bench. You measure the **potential difference** across a component and the **current** through it, and then calculate:

$$R = \frac{V}{I} \qquad \text{resistance } (\Omega) = \frac{\text{potential difference (V)}}{\text{current (A)}}$$

Why nichrome and not copper? That is a question about *resistivity* — see below.

Resistivity (background — not on the AQA specification)



$$R = \frac{\rho L}{A}$$

- ρ **Resistivity — belongs to the material.** The symbol is the Greek letter **rho**, written ρ — it is not a letter p. Copper and nichrome are different substances with different chemical properties, and that difference *is* ρ. It is why one wire is used for connecting leads and the other for heating elements.
- A **Cross-sectional area — how thick the wire is.** This is the one we change all the time in practice: different *gauges* of wire, or simply different thicknesses.
- L **Length.** The one you are changing today.
- R **Resistance — the overall opposition to the current in that piece of wire**, which is what your meters let you work out.

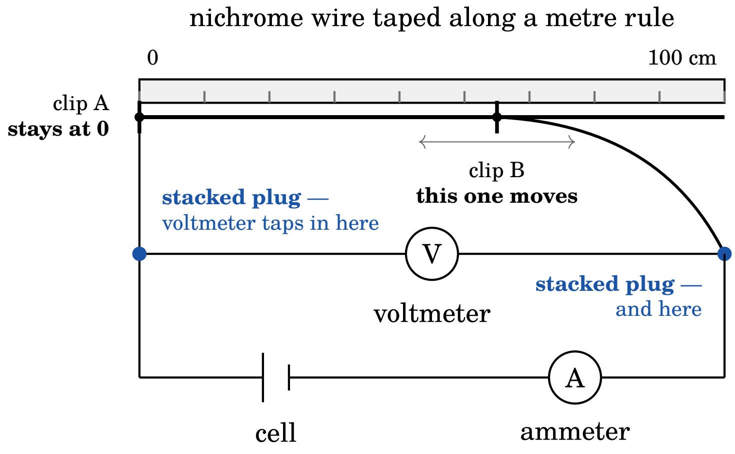
The question

How does the resistance of a wire depend on its length?

Before you touch the apparatus, decide what you are changing, what you are measuring, and what you must hold still.

Independent variable what you change	*The length of the nichrome wire between the crocodile clips.*
Dependent variable what you measure	*The resistance of that length of wire — found from V and I.*
Control variables what stays the same	*The same wire throughout (same material, same thickness / diameter); the same supply setting; the temperature of the wire — keep the current low and disconnect between readings.*

The circuit



The ammeter sits in the loop, so the whole current goes through it. The voltmeter sits across *the stretch of wire between the two clips — and nowhere else. The length you are testing is the length between clip A and clip B.*

Method

1. Build the circuit above. Leave **one lead off the cell** until you are checked — that gap is your switch.
2. **At each blue junction on the diagram, use two leads with stacked banana plugs** — push one plug into the back of the other rather than using a single long lead. This does two jobs at once: it gives you enough reach for the 100 cm readings, and it leaves a free socket for the voltmeter to plug straight into.
3. Attach **clip A at the 0 cm mark** and leave it there for the whole experiment. Every length you record is measured from this clip.
4. Move **clip B** to **100 cm** — the far end. You are starting with the *longest* piece of wire and working inwards.
5. Connect the last lead. Read the **ammeter** and the **voltmeter**. Record both. **Disconnect that lead again straight away.**
6. Move clip B in to 90 cm and repeat. Then 80, 70, 60, 50, 40 and **stop at 30 cm**.
7. Calculate the resistance of each length using $R = V/I$. Give your answers to **2 decimal places**.

Safety and good practice

- **The wire gets hot.** Do not leave the circuit connected between readings — take the reading, then pull the lead off the cell.
- Do not touch the wire while the circuit is connected.
- If the wire starts to glow or smell, pull the lead off and put your hand up.
- **Keep the wire taut against the rule.** A sagging wire between the clips is *longer* than the length you read off the rule, and every reading after it is wrong.

Why we start long and stop at 30 cm

It is a chain, and each link causes the next:

short wire → small resistance → large current → the wire heats up → its resistance changes

A very short piece has so little resistance that the current through it becomes large, and a large current heats the wire. Once the wire is hot its resistance is no longer the resistance you set out to measure, and those readings are worthless. Below about 30 cm the effect is bad enough to ruin the graph — so we do not go there.

Results

Length / cm	Potential difference / V	Current / A	Resistance / Ω
100			
90			
80			
70			
60			
50			
40			
30			

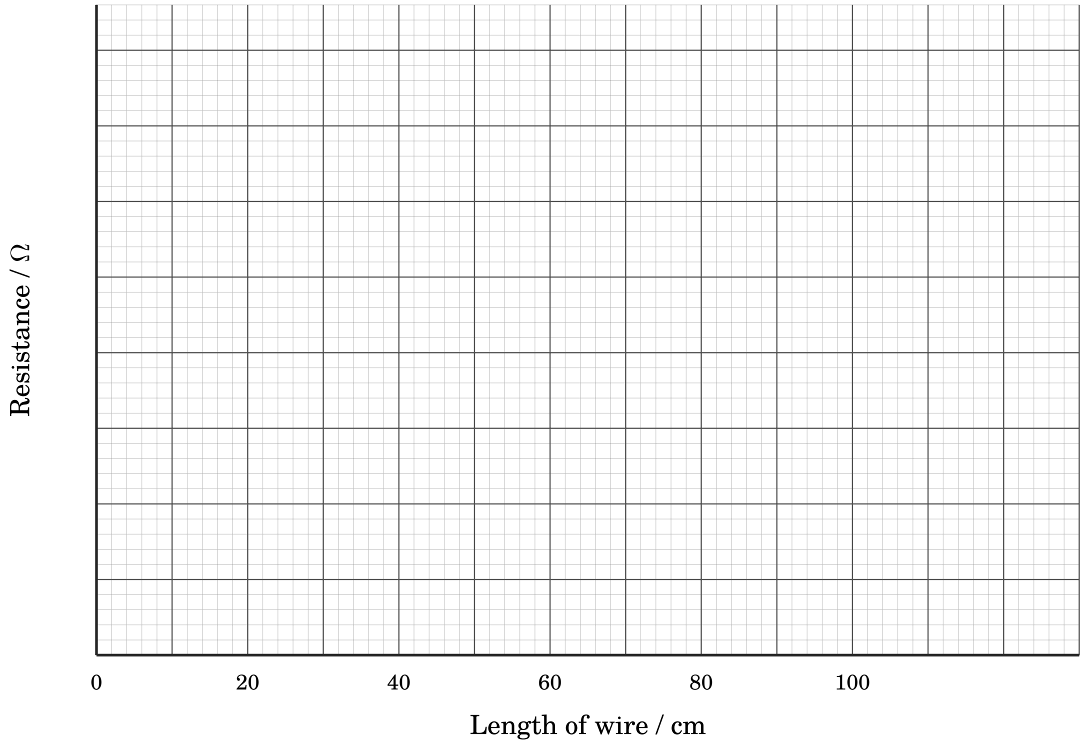
Show one resistance calculation in full, so I can see the working.

Example, using 100 cm: $V = 4.71$ *V,* $I = 0.08$ *A, so* $R = V/I = 4.71/0.08 = 58.88\ \Omega$*. (Students' own figures will differ — the method is what is being marked.)*

The graph

Plot **resistance** on the vertical axis against **length** on the horizontal axis. Choose your own scale for resistance so the points fill the grid. Draw a **line of best fit** — a single straight line, not dot-to-dot.

$(0, 0)$ **is not a data point.** Fit the line to your readings only. Do not force it through the origin.



What the graph tells you

1) **Describe** the shape of your graph in one sentence.

A straight line with a positive gradient. Extended backwards it passes through, or close to, the origin — though note we have no readings below 30 cm, so that part of the line is extrapolated.

2) **Circle** one to complete the conclusion: **the resistance of the wire is ____________ to its length.**

directly proportional   indirectly proportional

inversely proportional   exponentially proportional

3) **Use** your line of best fit to predict the resistance of a **75 cm** length. Show on the graph how you got it.

Read up from 75 cm to the line, then across to the resistance axis. Answer depends on their own gradient; the method — drawn construction lines on the graph — is the point.

4) **Explain** why a longer wire has more resistance, in terms of what is happening *inside* the wire.

The current is a flow of electrons through the metal. A longer wire contains more positive metal ions in the way, so there are more collisions between the electrons and the ions. More collisions means the flow is opposed more, so the resistance is greater.

Evaluation

5) Give one reason why your line of best fit may not pass exactly through the origin, even though it should.

It is very hard to place crocodile clip A exactly on the 0 cm mark, so every length reading carries the same small error (a zero error). There is also some resistance in the leads and at the contacts of the clips themselves, which adds a small fixed amount to every reading.

6) Suggest one improvement for a student who takes only one reading at each length, and say what it would give them.

Repeat each length two or three times and take a mean. This reduces the effect of random error — a single misread meter or a badly placed clip — and lets them spot an anomalous result rather than plotting it.

7) Predict what happens to the resistance if another student uses a *thicker* nichrome wire of the same length, and explain why.

The resistance decreases. A thicker wire gives the electrons a larger cross-section to flow through — more routes side by side — so for the same potential difference a larger current flows, which means a lower resistance.

Extra space for working

Rough calculations, a second attempt at a graph reading, anything that did not fit.

Before you pack away: disconnect at the cell, unclip the wire, and check it is *cool* before you coil it. Meters back in the tray the way you found them.